18.3 Wave-particle duality and the De Broglie wavelength

Planck's and Einstein's work established beyond doubt that electromagnetic radiation must have a particle nature in order to explain certain experimental observations, such as the photoelectric effect [5]. This is reminiscent of Newton's corpuscular theory of light, although the two are not connected. However, interference and diffraction experiments showed that electromagnetic radiation can also behave like a wave. Hence, electromagnetic radiation has a dual character; it exhibits a wave-particle duality. In certain circumstances it behaves like a wave, while in other circumstances, it behaves as a beam of particles.

Bohr postulated the orthodox *complementarity principle* that made the particle character and the wave character mutually exclusive [6]. It asserted that in situations where the particle character is manifested, the wave character is completely suppressed and vice versa. This orthodox principle has since been disproved by a remarkable set of experiments. Ghose, Home and Agarwal [7] had proposed a biprism experiment schematically depicted in Fig. 18.2. A single photon source emits a single photon which is split into orthogonal states ψ_r and ψ_t by a 50:50 beam splitter. They are detected by two photon detectors D_r and D_t. If the photon behaves truly as a "particle," then it should be detected at either D_r or D_t (but never at both) since a particle cannot traverse two paths simultaneously. That is, there should be perfect anti-coincidence between D_r and D_t, or, in other words, *either* D_r *or* D_r will click but *both* will never click simultaneously in between the arrival of two successive photons. The clever twist in this experiment, motivated by an experiment performed in the 19th century by Jagadish Chandra Bose [8], is the placement of the biprism with a small tunneling gap in the path of the transmitted photon. If D_t clicks *and* D_r *does not*, then we have made a "which path" (*welcher weg*) determination (the particle took the path of transmission as opposed to reflection) and a sharp *particle nature* is demonstrated [9]. Yet, to arrive at D_t, the particle must have tunneled through the biprism and tunneling is a sharp *wave* attribute. Therefore, this gedanken experiment presents a conundrum: the photon is behaving as both a particle and a wave in the *same experiment*, disproving the strict Bohr Complementarity Principle.

This experiment was actually conducted by Mizobuchi and Ohtake [10], using a single photon source. Perfect anti-coincidence was found between D_t and D_r, demonstrating the particle nature. Yet, the very fact that D_t ever clicked required tunneling and hence the existence of a wave nature. It was claimed that, in this experiment, a photon was behaving both sharply as a particle and as a wave in violation of orthodox Bohr's Complementarity Principle. A slight modification of this experiment has been proposed by Rangwala and Roy [11] where interference is used instead of tunneling to showcase the wave-like behavior. They claimed that quantum mechanics does not prohibit the

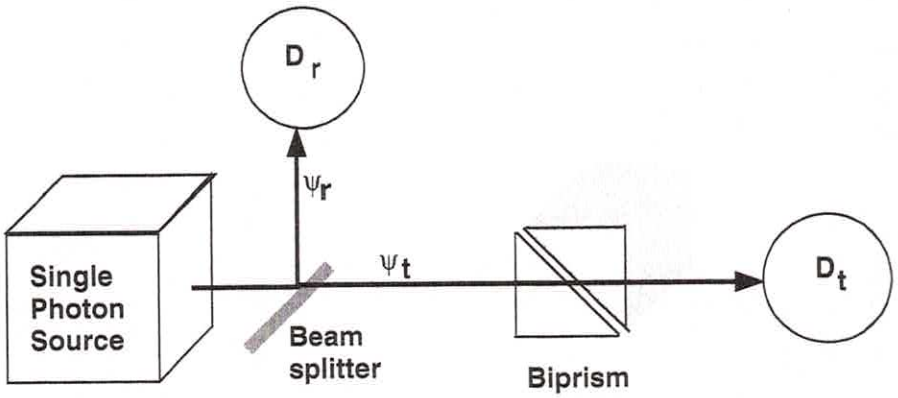

FIGURE 18.2
An experiment to disprove the orthodox Bohr's complementarity principle.

demonstration of simultaneous wave and particle behavior; rather, it prohibits their simultaneity only when the wave and particle attributes are "complementary" [12] in the sense that projection operators associated with them do not *commute*. Their view is that in the experiments of [7, 10, 11], the wave and particle behavior supposedly are not truly complementary and hence not subject to the restrictions of the orthodox Complementarity Principle.

A somewhat tempered version of the Complementarity Principle is the *duality principle* due to Englert [13] which states that a quantum system can simultaneously exhibit wave and particle behavior, but sharpening of the wave character blurs the particle character and vice versa. This is an extension of earlier work by Wootters and Zurek [14], Greenberger and Yasin [15], and Mandel [16]. In fact, Englert derives an inequality

$$P^2 + V^2 \leq 1, \tag{18.13}$$

where P is a measure of the particle attribute and V is a measure of the wave attribute. Equation (18.13) immediately shows that stronger wave or stronger particle behavior can be manifested only at the expense of each other.

Returning to the wave-particle duality, Louis de Broglie made another breakthrough in 1924 by postulating that if electromagnetic radiation could have a dual wave-particle character, then material objects will also behave like waves. In other words, de Broglie posited that, if a particle passes through a slit whose width is comparable to the wavelength associated with it, then it should undergo diffraction just as photons do in a single-slit experiment.

Louis de Broglie conjectured that the wavelength associated with a material body would satisfy the same relation as photons, a conjecture which Einstein considered brilliant. Therefore, a material body with a momentum p will have

a wavelength given by

However, there is one import
jects in the way their wave an
for a photon, only one relati
quency from a photon's par
massive object, on the other
its wavelength and frequency
(kinetic + rest mass) is relate

where $E_0 = m_0c^2$ and m_0 is t
in Chapter 2 to ultimately de
The total energy E is also

Therefore, according to de
tivistic particle must be deter

$$\lambda_r = \frac{hc}{pc} =$$

Exercises:

- What is the accelerating
wavelength of 5 Å, the s
monly studied semicond

Solution

For an electron (with a
potential difference ΔV,

$$K =$$

For $\lambda = 5$ Åand m_0 equal
is $\Delta V \sim 6$ V.

- Calculate the de Broglie
ergy equal to the longitu
in GaAs. The effective
$0.067m_o$, m_o being the fr